

Exam 1 Solution Manual 1 DE

Owen Stayton

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1 Problem 1

Solve the DE

$$\frac{dy}{dx} = \frac{xe^{x^2}}{y(y + \ln(y))}$$

1.1 Solution

We will use Separation of Variables to solve this DE

$$(y + \ln(y))dy = xe^{x^2}dx$$

Now integrate

$$\int y^2 dy + \int y \ln(y) dy = \int xe^{x^2} dx$$

Let's isolate to solve each integral

$$\int y \ln(y) dy = \frac{y^2 \ln(y)}{2} - \int \frac{y^2}{2y} dy$$

$$\int y \ln(y) dy = \frac{y^2 \ln(y)}{2} - \frac{y^2}{4} + K_1$$

$$\int y^2 dy = \frac{y^3}{3} + K_2$$

$$\int xe^{x^2} dx = \frac{e^{x^2}}{2} + K_3$$

Now plug back into equation

$$\frac{y^2 \ln(y)}{2} - \frac{y^2}{4} + K_1 + \frac{y^3}{3} + K_2 = \frac{e^{x^2}}{2} + K_3$$

Final Answer

$$4y^3 + 6y^2 \ln(y) - 3y^2 = 6e^{x^2} + K_4$$

2 Problem 2

Solve the DE

$$\frac{dy}{dx} = \frac{2y^4 + x^4}{xy^3}$$

2.1 Solution

Notice that the equation has the form of a homogeneous first-order equation:

$$\frac{dy}{dx} = f(x, y), \quad \frac{dy}{dx} = \frac{2y^4 + x^4}{xy^3}$$

It's not clear if we could easily solve this using separation of variables. Let's make the substitution then solve like a separable equation:

$$\begin{aligned} y &= xv, & \frac{dy}{dx} &= v + x \frac{dv}{dx} \\ v + \frac{dv}{dx}x &= \frac{2x^4v^4 + x^4}{x^4v^3} \\ x^4v^4 + \frac{dv}{dx}x^5v^3 &= 2x^4v^4 + x^4 \\ \frac{dv}{dx}x^5v^3 &= x^4v^4 + x^4 \\ \frac{dv}{dx}x^5v^3 &= x^4(v^4 + 1) \\ \frac{v^3}{v^4 + 1}dv &= \frac{x^4}{x^5}dx \end{aligned}$$

Now integrate

$$\int \frac{v^3}{v^4 + 1} dv = \int \frac{x^4}{x^5} dx$$

Isolate and solve each integral

$$\begin{aligned} \int \frac{v^3}{v^4 + 1} dv &= \frac{1}{4} \ln(v^4 + 1) + K_1 \\ \int \frac{x^4}{x^5} dx &= \frac{1}{5} \ln|x^5| + K_2 \end{aligned}$$

Plug back in

$$\frac{1}{4} \ln(v^4 + 1) + K_1 = \frac{1}{5} \ln|x^5| + K_2$$

Get rid of the logs

$$\exp[\ln((v^4 + 1)^{1/4})] = \exp[K_3 + \ln|x|]$$

$$(v^4 + 1)^{1/4} = xK_4$$

Make the substitution

$$v = \frac{y}{x}$$

$$\left(\frac{y}{x}\right)^4 + 1)^{1/4} = xK_4$$

Now solve for y^4

$$y^4 = x^8 K_5 - x^4$$

3 Problem 3

Solve the DE

$$(2x^2y - 2x^3)dy + (4x^3 - 6x^2y + 2xy^2)dx = 0$$

3.1 Solution

Notice this problem has the form of an exact DE

$$M(x, y)dx + N(x, y)dy = 0, \quad (2x^2y - 2x^3)dy + (4x^3 - 6x^2y + 2xy^2)dx = 0$$

Let's test for exactness:

$$\frac{\partial M}{\partial y} = 4xy - 6x^2, \quad \frac{\partial N}{\partial x} = -6x^2 + 4xy$$

This equation is exact since M_y and N_x are equivalent. Take M and integrate with respect to x

$$\int M(x, y)dx = \int (4x^3 - 6x^2y + 2xy^2)dx$$

$$F = x^4 - 2x^3y + x^2y^2 + f(y)$$

Now differentiate with respect to y

$$\frac{\partial F}{\partial y} = -2x^3 + 2x^2y + f'(y)$$

Now set $\partial F_y = N$

$$-2x^3 + 2x^2y + f'(y) = 2x^2y - 2x^3$$

$$f'(y) = 0$$

Integrate with respect to y and plug into $F(x, y)$

$$F = x^4 - 2x^3y + x^2y^2 + K_1$$

4 Problem 4

$$\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{x}{1+x^2}$$

4.1 Solution

Notice this problem has the same form as a first-order linear DE

$$\frac{dy}{dx} + p(x)y = q(x), \quad \frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{x}{1+x^2}$$

Find the integrating factor

$$IF = \exp\left[\int \frac{2x}{1+x^2}dx\right]$$

$$IF = 1 + x^2$$

$$\int d((1+x^2)y) = \int (x)dx$$

$$y = \frac{x^2 + K_2}{2(1+x^2)}$$

5 Problem 5

A 200-gallon tank initially contains 80 gallons of pure water. Starting at $t = 0$, salt water containing 2.0 pounds of slater per gallon falls into the the tank at 5 gal/min. The thoroughly mixed solution flows out at 3 gal/min. How much salt is in the tank when the tank overflows?

5.1 Solution

$$\frac{dx}{dt} = 10 - \frac{3x}{80+2t}$$

DE is linear, find IF

$$IF = (80+2t)^{3/2}$$

$$\int d(x(80+2t)^{3/2}) = \int 10(80+2t)^{3/2}dt$$

$$x(80+2t)^{3/2} = 2(80+2t)^{5/2} + K_1$$

$$K_1 = -2(80)^{5/2}$$

Find t when the tank overflows

$$200 = 80 + 2t$$

$$t = 60$$

Plug into equation and solve

$$x = 2(80 + 120) - \frac{2(80)^{5/2}}{(80 + 120)^{3/2}}$$

6 Problem 6

At $t = 0$, a 48-pound object is pushed beneath the surface of a calm lake and then released from rest. The object has a buoyancy force of 57 pounds, so it begins to rise after release. The resistance of the water (in pounds) is given by $\frac{v^2}{4}$, where v is the upward velocity of the object in feet per second. If the object reaches the surface in 3 seconds, what is the velocity of the object when it reaches the surface of the lake?

6.1 Solution

$$57 - 48 - \frac{v^2}{4} = \frac{48}{32} \frac{dv}{dt}$$

Do the algebra to set it up as separation of variables problem

$$\frac{-1}{6} dt = \frac{1}{v^2 - 36} dv$$

$$\int \frac{-1}{6} dt = \int \frac{1}{v^2 - 36} dv$$

Split the integral

$$\int \frac{-1}{6} dt = \frac{-t}{6}$$

$$\int \frac{1}{v^2 - 36} dv = \int \frac{A}{v - 6} + \frac{B}{v + 6} dv$$

$$A = \frac{1}{12}, \quad B = \frac{-1}{12}$$

$$\int \frac{1}{12(v - 6)} + \frac{-1}{12(v + 6)} dv = \frac{1}{12} \ln \left| \frac{v - 6}{v + 6} \right|$$

$$\frac{-t}{6} = \frac{1}{12} \ln \left| \frac{v - 6}{v + 6} \right| + K_1$$

$$-2t = \ln \left| \frac{v - 6}{v + 6} \right| + K_2$$

Get rid of constant

$$-2(0) = \ln|-1| + K_2$$

$$K_2 = \ln|-1^{-1}| = 0$$

Solve for v

$$v = \frac{6(e^{-2t} + 1)}{1 - e^{-2t}}$$

Solve for v when $t = 3$

$$v = \frac{6(e^{-6} + 1)}{1 - e^{-6}}$$

7 Problem 7

He combined weight of a boy and his sled is 120 pounds. Starting from rest at $t = 0$, he sleds down a hill whose slope is $\frac{3}{4}$. The air resistance (in pounds) is numerically equal to twice the velocity (in feet/sec) of the boy and his sled. If the coefficient of sliding friction between the sled's runners and the sloping hill is $\frac{1}{12}$, (a) what is the velocity of the sled at the end of t seconds, and (b) how long (correct to the nearest foot) is the hill if it take 6 seconds to reach the bottom of the hill?

7.1 Solution

$$W = 120, W \sin(\theta) = 72, W \cos(\theta) = 96, \mu N = 8$$

$$F_{net} = 72 - (8 + 2v), \quad m = \frac{120}{32}$$

$$\frac{15}{4} \frac{dv}{dt} = 64 - 2v$$

Do the algebra to solve as linear DE

$$\frac{dv}{dt} + \frac{8v}{15} = \frac{256}{15}$$

$$IF = e^{\int \frac{8}{15} dt} = e^{8t/15}$$

$$\int d(e^{8t/15} v) = \int \frac{256}{15} e^{8t/15} dt$$

$$e^{8t/15} v = 32 e^{8t/15} + K_1$$

$$v = 32 + K_1 e^{-8t/15}$$

Get rid of K_1

$$K_1 = -32$$

Solve as a function of position

$$\int dr = \int 32 - 32 e^{-8t/15} dt$$

$$r = 32t + 60e^{-8/15} + K_2$$

Get rid of K_2

$$K_2 = -60$$

Solve part a

$$v = 32 - 32e^{-48/15}$$

Solve part b

$$r = 32(6) + 60e^{-48/15} - 60$$